RESEARCH ROADMAP

A Data-Driven Analysis of Migration and Remittance:

Insights for Related Decision-Making in Sri Lanka (2000–2025)

Discipline: Data Science & Analytics | Component: Machine Learning
Dataset Period: 2000 – 2025 | 13 Features | 26 Annual Observations

1. Research Overview & Positioning

This roadmap outlines the complete analytical structure for a data science research paper examining Sri Lanka's labour emigration trends from 2000 to 2025. The study integrates descriptive analytics, inferential statistics, time-series modelling, and supervised machine learning to generate policy-relevant insights on emigration volume, workforce skill composition, gender dynamics, contract duration, and poverty linkage.

The dataset contains 13 variables spanning 26 annual observations, enabling both longitudinal trend analysis and predictive modelling. The sections below define each research area, the specific analytical questions it addresses, and the methods to employ.

2. Dataset Description & Feature Engineering

2.1 Variable Inventory

The dataset consists of the following 13 features:

Feature	Type	Description
year	Temporal (int)	Reference year — used as time index
emigration	Continuous	Total annual labour emigration count
male_perc / female_perc	Continuous (%)	Gender composition of emigrants
total_skilled_perc	Continuous (%)	Share of skilled labour in total emigration
total_lowskilled_perc	Continuous (%)	Share of low-skilled labour in total emigration
male_skilled_perc	Continuous (%)	Skilled share within male emigrants
male_lowskilled_perc	Continuous (%)	Low-skilled share within male emigrants

female_skilled_perc	Continuous (%)	Skilled share within female emigrants
female_lowskilled_perc	Continuous (%)	Low-skilled share within female emigrants
average_age_of_emigrant	Continuous	Mean age of outgoing workers
average_contract_years	Continuous	Average contracted employment duration
world_bank_poverty_less_than_3.65_dollars_per_day	Continuous (%)	Population below $3.65/day poverty line (WB)

2.2 Feature Engineering

Prior to analysis, derive the following computed features to enrich the dataset and expand modelling capacity:

- Emigration Growth Rate (YoY %): annual percentage change in emigration volume to capture momentum and shock events
- Skill Gap Index: difference between total_skilled_perc and total_lowskilled_perc as a single composite indicator
- Gender Parity Index: ratio of female_perc to male_perc to measure shifts in gender composition over time
- Skilled Female Ratio: female_skilled_perc as a proportion of total female emigration, tracking upskilling among women
- Poverty Reduction Rate: year-over-year change in the World Bank poverty variable
- Contract Duration Category: binned version of average_contract_years (short: <2.2, medium: 2.2–2.8, long: >2.8)
- Crisis Indicator: binary flag for years with extreme emigration shocks (e.g., 2020 COVID collapse)

3. Exploratory Data Analysis (EDA)

3.1 Descriptive Statistics & Distribution Analysis

Establish the statistical baseline of all variables before modelling.

- Compute mean, median, standard deviation, skewness, and kurtosis for all continuous variables
- Generate a correlation matrix (Pearson and Spearman) to identify multicollinear and inversely related features
- Plot histograms and box plots for emigration, poverty rate, and age to assess normality and detect outliers
- Identify structural anomalies: the 2020 COVID-19 shock (emigration collapsed from ~203K to ~54K) and the 2022 economic crisis spike (emigration jumped to ~311K)

3.2 Temporal Trend Visualisation

Examine how each variable evolves over the 2000–2025 window using time-series line plots, dual-axis charts, and annotated event markers.

- Overall emigration volume trend with recession and crisis annotations
- Gender composition shift: female majority (67%) in 2000 eroding to male majority by 2013 — visualise the crossover point
- Skill level evolution: decline from ~80% skilled in early 2000s to ~52% in 2018–2019, then recovery to ~77% by 2024–2025
- Contract duration elongation: steady increase from 2.0 years (2000) to 3.5 years (2025)
- Poverty trajectory: near-monotonic decline from 58.2% (2000) to 11.3% (2019), then rebound to 27.1% (2023) post-crisis

3.3 Bivariate & Multivariate Relationships

- Scatter plots: emigration vs. poverty rate; poverty rate vs. skilled_perc; contract years vs. age
- Pair plot of all continuous features to surface unexpected correlations
- Heatmap of the full correlation matrix with significance thresholds annotated
- Cross-tabulation analysis: skill level by gender across three era segments (2000–2009, 2010–2019, 2020–2025)

4. Temporal Analysis & Trend Decomposition

4.1 Time-Series Decomposition

Decompose emigration and the poverty rate into trend, seasonality (limited here to annual cycles), and residual components using classical decomposition and STL (Seasonal-Trend decomposition using LOESS). Identify structural breakpoints using the Bai-Perron test or CUSUM analysis, expected around 2008 (global financial crisis), 2013 (male majority inflection), 2018–2019 (skill ratio collapse), and 2020 (COVID shock).

4.2 Rolling Statistics & Volatility

- Compute 3-year and 5-year rolling means and standard deviations for emigration and skill ratios
- Identify high-volatility sub-periods to contextualise ML model uncertainty
- Calculate coefficient of variation (CV) per decade to compare stability across eras

4.3 Cross-Correlation Analysis (CCF)

Examine whether changes in emigration volume lead or lag changes in the poverty rate using Cross-Correlation Function (CCF) plots. A positive lag would support the remittance-to-poverty-reduction hypothesis. Test lags from t-1 to t-5 years.

5. Gender Dynamics in Labour Emigration

5.1 Longitudinal Gender Composition Analysis

This section specifically addresses the structural gender reversal observed in the dataset — from female-dominant emigration in 2000 to male-dominant by 2013 — and its workforce implications.

- Plot the male/female percentage crossover with policy event overlays (e.g., Gulf Cooperation Council domestic worker bans on Sri Lankan women issued in 2013)
- Measure the rate of gender composition change per year using first-order differencing
- Compare gender parity index across sub-periods with ANOVA or Kruskal-Wallis test

5.2 Gender-Differentiated Skill Analysis

Disaggregate the skill analysis by gender to test whether female and male emigrants follow different skill trajectories.

- Line charts of female_skilled_perc vs. male_skilled_perc over time
- Note the extreme female skill dominance in early years (female_skilled_perc ~97% in 2005) and its decline to ~71% by 2022 during the economic crisis
- Pearson correlation between female_perc and female_skilled_perc — test if gender composition shifts track with skill degradation

5.3 Gender & Contract Duration

- Box plots of average_contract_years across gender-dominant periods (female-dominant pre-2013 vs. male-dominant post-2013)
- Correlation between gender composition ratio and average_contract_years

6. Skill Composition & Workforce Quality Analysis

6.1 Skilled vs. Low-Skilled Emigration Trends

The dataset reveals a non-monotonic pattern in skill composition — broad skilled dominance, a sharp contraction in 2018–2019, and partial recovery. Analysing this trajectory is central to Sri Lanka's human capital policy debate.

- Plot total_skilled_perc and total_lowskilled_perc as stacked area charts over 2000–2025
- Identify the skill inversion event of 2018–2019 where low-skilled proportion approached parity (~47.6%)
- Compute the Skill Gap Index (derived feature) and examine its correlation with the poverty rate

6.2 Brain Drain Assessment

Evaluate whether rising emigration is associated with deteriorating or improving skill composition — a core brain drain hypothesis.

- Scatter plot of emigration volume vs. total_skilled_perc across years, testing whether high-emigration years are high-skill or low-skill years
- Compute Spearman rank correlation between emigration and skilled percentage

- Segment analysis: do crisis-year spikes (2022, 2024–2025) differ in skill composition from pre-crisis emigration?

6.3 Age & Contract Duration as Skill Proxies

- Analyse whether average_age_of_emigrant trends correlate with skill ratios — older emigrants may reflect more experienced, skilled workers
- Examine contract duration elongation (2.0 to 3.5 years over 25 years) as a signal of rising employer demand for more specialised workers
- Scatter matrix: age vs. contract years vs. skilled percentage

7. Migration–Poverty & Remittance Nexus

7.1 Emigration–Poverty Linkage

Test the hypothesis that increased emigration — and inferred remittances — correlates with poverty reduction in Sri Lanka. This is the primary policy-relevant relationship in the study.

- Overlay emigration volume and poverty rate on a dual-axis time-series chart
- Compute lagged correlations (CCF) to determine whether emigration precedes poverty reduction by 1–3 years
- Granger causality test: does emigration Granger-cause poverty reduction? (test at 1, 2, 3 lag orders)

7.2 Skill Composition & Poverty

Investigate whether the skill level of emigrants — a proxy for remittance earnings — moderates the poverty impact.

- Scatter plot of total_skilled_perc vs. poverty rate with year labels
- Partial correlation between skilled percentage and poverty rate, controlling for emigration volume
- Compare poverty reduction rate during high-skill vs. low-skill emigration periods

7.3 Post-Crisis Poverty Rebound Analysis

The poverty rate reversed from 11.3% in 2019 to 27.1% in 2023 following the economic crisis. Analyse whether emigration patterns during 2020–2023 amplified or cushioned this rebound.

- Isolate the 2020–2023 sub-window and examine emigration, skill, and poverty co-movements
- Compute the elasticity of poverty with respect to emigration volume in this period

8. Machine Learning Component

8.1 Task 1 — Emigration Volume Forecasting (Regression)

Objective: Predict future annual emigration volume using historical features.

Component	Methods / Tools	Variables Used
Target Variable	emigration	Continuous
Features	All 12 remaining variables + engineered features	Year, gender %, skill %, age, contract yrs, poverty
Baseline Models	Linear Regression, Ridge, Lasso	Regularisation to handle multicollinearity
Advanced Models	Random Forest Regressor, Gradient Boosting (XGBoost)	Capture non-linear interactions
Validation	Time-series cross-validation (expanding window)	Prevent data leakage from future to past
Metrics	MAE, RMSE, R^2, MAPE	Interpretable error in emigration count units

8.2 Task 2 — Poverty Rate Forecasting (Regression)

Objective: Forecast the World Bank poverty rate using emigration and workforce composition features as predictors — directly testing the emigration-remittance-poverty pathway.

Component	Methods / Tools	Variables Used
Target Variable	world_bank_poverty_less_than_3.65_dollars_per_day	Continuous (%)
Features	emigration, skilled_perc, gender_perc, contract_years, age	Lagged features at t-1 and t-2
Models	VAR (Vector Autoregression), Random Forest, SVR	VAR for multivariate dynamics; RF/SVR for non-linearity
Key Test	Include/exclude lagged emigration to measure marginal contribution	Isolates the emigration effect on poverty
Metrics	MAE, RMSE, Theil's U statistic	Theil's U benchmarks against naive forecast

8.3 Task 3 — Emigration Regime Classification (Classification)

Objective: Classify each year into a meaningful emigration regime to support scenario-based policy planning.

- Define three regime classes based on emigration volume and skill composition:
  - Class 0 — Contraction: emigration below 200K (e.g., 2006, 2017–2019, 2020–2021)
  - Class 1 — Moderate Growth: 200K–280K (e.g., 2000–2012)
  - Class 2 — High Surge: above 280K (e.g., 2014, 2022–2025)

Component	Methods / Tools	Variables Used

Target	Regime class (0/1/2)	Derived from emigration volume + skill split
Features	All engineered features + year trends	Including crisis indicator flag
Models	Logistic Regression, Decision Tree, Random Forest	Decision Tree for interpretability; RF for performance
Validation	Leave-One-Out CV (LOOCV)	Preferred for small N=26 dataset
Metrics	Accuracy, Macro F1, Confusion Matrix	Macro F1 for class-balanced evaluation

8.4 Feature Importance & Interpretability

After fitting the ML models, extract and visualise feature importance to directly connect analytical findings to policy conclusions.

- Random Forest: permutation importance and mean decrease impurity (MDI)
- Gradient Boosting: SHAP (SHapley Additive exPlanations) values to show per-observation feature contributions
- Lasso: non-zero coefficients as a built-in feature selector
- Produce a ranked feature importance chart for the primary emigration forecasting model

9. Statistical Inference & Hypothesis Testing

9.1 Hypothesis Tests to Include

The following formal hypothesis tests ground the descriptive findings in statistical evidence:

Hypothesis	Test	Variables
Higher emigration correlates with lower poverty (with lag)	**Granger Causality / CCF**	emigration (t-k), poverty (t)
Skilled emigration % differs significantly across eras	**ANOVA / Kruskal-Wallis**	total_skilled_perc across 3 periods
Male vs. female skill levels differ significantly	**Independent t-test / Mann-Whitney U**	male_skilled_perc, female_skilled_perc
Gender composition changed structurally around 2013	**Chow Structural Break Test**	male_perc, female_perc series
Average contract duration has a significant upward trend	**Mann-Kendall Trend Test**	average_contract_years
Emigration series is non-stationary	**Augmented Dickey-Fuller (ADF) Test**	emigration

10. Policy Insights & Decision-Making Framework

10.1 Evidence-Based Policy Recommendations

Translate analytical findings into concrete, stakeholder-targeted recommendations. Structure them by decision-maker type:

- Ministry of Labour & Foreign Employment: skill composition data supports upskilling intervention priorities
- Ministry of Finance: emigration-poverty lag analysis informs the timing of remittance-linked social protection policies
- Central Bank of Sri Lanka: emigration volume forecasts support remittance inflow projections for forex reserve planning
- Women's Bureau: gender composition decline among female emigrants warrants targeted female worker protection policy

10.2 Scenario Analysis Using ML Forecasts

Using the trained regression models, construct three emigration scenarios for 2026–2028 and project their expected poverty impact:

- Scenario A (Optimistic): continued high emigration at ~310K with improving skill ratios
- Scenario B (Baseline): stabilisation around 280K with current skill composition
- Scenario C (Pessimistic): emigration contraction below 200K following a hypothetical external demand shock

11. Recommended Paper Structure

Section	Title	Maps to Roadmap Section
I	**Introduction & Research Questions**	Background, 5 research questions, dataset overview
II	**Literature Review**	Remittance–poverty nexus; brain drain; gender & migration
III	**Data & Methodology**	Sections 2 & 8 — dataset, feature engineering, ML task definitions
IV	**Exploratory Data Analysis**	Sections 3 & 4 — descriptive stats, trends, decomposition
V	**Gender Dynamics Analysis**	Section 5 — gender composition, skill-gender intersection
VI	**Skill Composition Analysis**	Section 6 — skill trends, brain drain, contract & age proxies
VII	**Migration–Poverty Nexus**	Section 7 — CCF, Granger tests, poverty linkage
VIII	**Machine Learning Results**	Section 8 — forecasting, classification, SHAP importance
IX	**Statistical Inference**	Section 9 — hypothesis test results & interpretation
X	**Policy Implications**	Section 10 — recommendations & scenario projections
XI	**Conclusion & Limitations**	Key findings, small-N caveat, future work

12. Recommended Tools & Libraries

12.1 Python Stack

- Data manipulation: pandas, numpy
- Visualisation: matplotlib, seaborn, plotly (interactive dashboards)
- Statistical testing: scipy.stats, statsmodels (Granger causality, ADF, VAR, ARIMA)
- Machine learning: scikit-learn (Random Forest, Logistic Regression, Ridge/Lasso, SVR, cross-validation)
- Gradient Boosting: xgboost or lightgbm
- Model interpretability: shap
- Time-series: statsmodels.tsa, pmdarima (auto-ARIMA)

12.2 Reporting & Reproducibility

- Jupyter Notebooks with clear section headers mapping to paper sections
- requirements.txt or environment.yml for reproducibility
- All figures exported at 300 DPI for publication quality
- GitHub repository with commit history demonstrating iterative development

13. Limitations & Mitigation Strategies

Explicitly acknowledging limitations strengthens the academic credibility of the paper:

Limitation	Mitigation
Small sample (N=26 annual observations)	Use LOOCV, bootstrap resampling, and regularised models; report confidence intervals
No actual remittance data — poverty linkage is inferential	Explicitly frame as a proxy analysis; cite World Bank remittance data in literature review
Aggregate data — no individual/household level granularity	Acknowledge ecological fallacy risk; recommend micro-level future work
Potential omitted variable bias (e.g., global oil prices, GCC demand)	Include limitations section; supplement with contextual narrative on external drivers
2025 data may be preliminary estimates	Flag 2025 as provisional; conduct sensitivity analysis with and without 2025

This roadmap is structured for journal/conference submission in data science, development economics, or migration studies. Follow Section 11 paper structure for the final manuscript.